

Assessing Fire Damage and Severity Using Machine Learning and Deep Learning

DATA 607

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Introduction

In 2024, Alberta experienced a severe wildfire season, with 1,230 recorded fires and approximately 714,874 hectares burned across the province. Among these events, the 2024 Jasper Wildfire Complex was the largest fire in Jasper National Park in over a century, burning approximately 35,000 hectares and destroying nearly half of the town (Parks Canada, 2025). More broadly, wildfires in Alberta have been increasing in both frequency and severity, driven by changing climatic conditions and fuel dynamics.

This trend underscores the urgent need for rapid and accurate methods to assess burned extent and vegetation loss. Satellite remote sensing offers consistent, large-scale spatial coverage, making it a powerful tool for wildfire monitoring. However, translating raw satellite imagery into actionable information requires robust analytical approaches capable of handling complex spectral and spatial patterns.

In this context, machine learning and deep learning methods provide promising solutions for automating burned area detection and improving classification accuracy. This project focuses on analyzing the extent and impact of the 2024 Jasper wildfire using satellite imagery in combination with these advanced computational techniques.

The primary objective of this study is to evaluate how machine learning and deep learning models can be integrated with satellite imagery to (1) detect the spatial extent of burned areas and (2) classify burn severity. Burned area detection is based on widely used spectral indices, including the Normalized Difference Vegetation Index, Normalized Burn Ratio, and the differenced NBR (dNBR), which capture vegetation health and fire-induced changes over time.

Data

Data Sources

The analysis utilizes multispectral imagery acquired from the Sentinel-2 platform, along with ancillary datasets obtained from the Alberta Open Data Portal. The selected imagery includes three key spectral bands relevant for burn analysis: Red, Near-Infrared (NIR), and Shortwave Infrared (SWIR).

Permission/Accessibility

All datasets used in this study are publicly available and free to access, provided appropriate attribution is given to the original sources.

Time Frame of Data Collection

Satellite images were acquired for both pre-fire and post-fire conditions to enable change detection. Pre-fire imagery was collected between May and June 2024, while post-fire imagery was obtained between July and August 2024, capturing the immediate impact of the wildfire event.

Methodology

This study adopts a structured, multi-stage workflow to determine forest fire extent from satellite imagery by integrating geospatial preprocessing, traditional machine learning, and deep learning techniques. The overall pipeline is designed to ensure robust feature extraction, accurate classification of burned areas, and reliable estimation of fire impact metrics.

1. Data Preprocessing

Satellite imagery is first prepared through a sequence of geospatial and radiometric processing steps. Multispectral raster datasets from Sentinel 2 are loaded and clipped to the defined Area of Interest (AOI), corresponding to the fire-affected region.

Vegetation and burn-sensitive spectral indices are computed, including the Normalized Difference Vegetation Index (NDVI) and the Normalized Burn Ratio (NBR):

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

$$NBR = \frac{(NIR - SWIR)}{(NIR + SWIR)}$$

Both indices are calculated for pre-fire and post-fire imagery. The differenced Normalized Burn Ratio (dNBR) is then derived as:

$$dNBR = NBR_{pre} - NBR_{post}$$

All derived indices, along with selected spectral bands (e.g., Red, NIR, SWIR), are stacked into a multi-layer feature space. Ground truth labels for burned and unburned areas are generated using a combination of threshold-based dNBR classification and, where available, reference fire perimeter datasets.

2. Machine Learning Models

Two supervised machine learning algorithms are implemented to classify burned versus unburned pixels:

Logistic Regression (Baseline Model):

A logistic regression classifier is used as a baseline to establish a linear decision boundary between classes. This model provides interpretability and serves as a benchmark for evaluating more complex approaches.

Random Forest (Primary Model):

A Random Forest classifier is employed as the primary model due to its robustness to noise, ability to capture nonlinear relationships, and effectiveness with high-dimensional data. The model consists of an ensemble of decision trees trained on bootstrapped samples, with feature randomness introduced at each split to reduce overfitting. Hyperparameters such as the number of trees, maximum depth, and minimum samples per leaf are optimized using cross-validation. Model performance is evaluated using accuracy, F1-score, and Cohen's Kappa.

3. Deep Learning Approach

To capture spatial context and improve classification performance, a Convolutional Neural Network (CNN) is developed for patch-based image classification. The CNN operates on 9x9 patches extracted from the stacked feature layers.

The CNN architecture consists of multiple convolutional layers with ReLU activation, followed by pooling layers to reduce spatial dimensionality, and fully connected layers for classification. The final output layer uses a sigmoid activation function to predict burned versus unburned classes.

4. Feature Engineering

The feature set integrates both spectral and index-based variables to enhance discriminative power:

- Pre-fire NDVI and post-fire NDVI (vegetation condition)
- Pre-fire NBR and post-fire NBR (burn sensitivity)
- dNBR (primary burn severity indicator)

These features capture both absolute spectral responses changes over time, enabling the models to distinguish burned areas from unaffected vegetation and other land cover types.

5. Impact Assessment Metrics

The final classified outputs are used to quantify fire extent and severity. Burn severity is categorized using standard dNBR thresholds (unburned, low, moderate, and high severity). These classes are mapped spatially to produce severity distribution maps.

Total burned area is computed by summing the number of pixels classified as burned and converting to square kilometers based on spatial resolution. Additionally, the proportional distribution of severity classes is calculated to assess the overall ecological impact of the fire.

This integrated methodology ensures a comprehensive analysis by combining pixel-level spectral information, spatial context from deep learning, and standardized fire severity metrics for accurate and scalable forest fire assessment.

Exploratory Data Analysis

The dataset consists of images collected in three different wavelengths. Near Infrared (NIR), Shortwave Infrared (SWIR), and Red. The NIR and Red images are collected at a resolution of 10m per pixel, while

the SWIR images are collected at a resolution of 20m per pixel. The SWIR images were up-sampled to match the resolution of the NIR and Red images. An example of a NIR images is shown below.

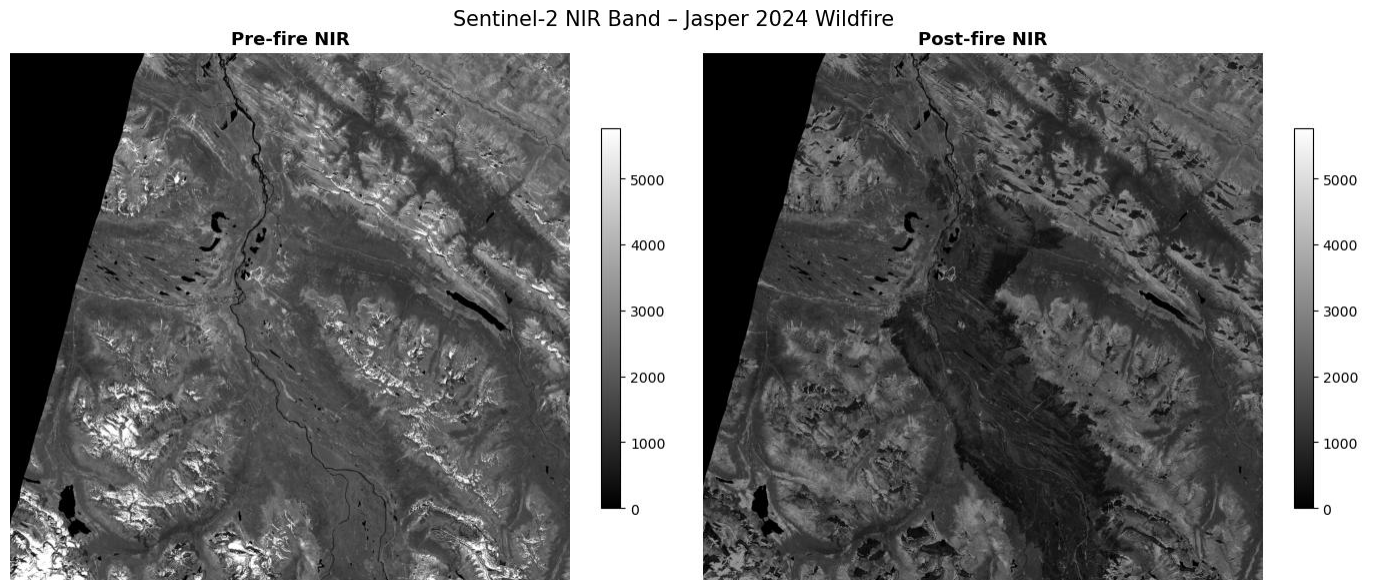


Figure 1: Pre- and Post-fire NIR Images

Reflectance in the range of 3000 to 4000 indicates dense and healthy vegetation. There is an abundance of healthy vegetation in the pre-fire image. The fire scarred area can be clearly seen in the post-fire image

Machine Learning Models

Train-test split

To ensure robust and unbiased model evaluation, a spatial data splitting strategy was employed instead of a conventional random split. In geospatial datasets derived from satellite imagery, neighboring pixels exhibit strong spatial autocorrelation, meaning that nearby observations tend to share similar spectral and class characteristics. Randomly partitioning such data can result in training and testing samples originating from the same geographic vicinity, leading to data leakage and overly optimistic performance estimates.

To mitigate this issue, the study area was divided into spatially contiguous tiles, which served as the basis for partitioning the dataset. These tiles were assigned to either the training or testing set in a mutually exclusive manner, ensuring that no geographic overlap existed between the two subsets. This block-based approach preserves spatial independence by preventing the model from learning location-specific patterns that would not generalize beyond the training region.

Additionally, a buffer zone was introduced between training and testing tiles to further reduce edge effects and minimize residual spatial dependence across boundaries. For the convolutional neural network (CNN) model, spatial consistency was maintained by ensuring that all image patches extracted from a given tile were assigned to the same data split.

The first 2915 columns (64%) of each image were used as the training set. The next 784 columns (16%) were used for validation, and the last 1075 columns (20%) were used for training.

Model Development and Evaluation

Four supervised learning models were developed and compared for the binary classification of burned versus unburned areas. All models were trained using a consistent feature set to ensure a fair comparison of their predictive capabilities. The input feature space consisted of eight variables derived from pre-fire and post-fire satellite imagery:

- Pre-fire Near-Infrared (NIR)
- Post-fire Near-Infrared (NIR)
- Pre-fire Shortwave Infrared (SWIR)
- Post-fire Shortwave Infrared (SWIR)
- Pre-fire Red band
- Post-fire Red band
- Pre-fire Normalized Difference Vegetation Index (NDVI)
- Post-fire Normalized Difference Vegetation Index (NDVI)

These features were selected to capture both spectral reflectance properties and vegetation dynamics before and after the fire event, which are critical for distinguishing burned areas from unburned land cover.

It is important to note that the class distribution was imbalanced across the dataset splits. The training and validation datasets each contained approximately 57% burned pixels, whereas the test dataset contained only 22% burned pixels. This discrepancy introduces a more challenging and realistic evaluation scenario, as the model must generalize to a distribution with fewer burned samples. However, it also necessitates the use of evaluation metrics beyond overall accuracy to properly assess model performance under class imbalance.

Model Descriptions

Logistic Regression

Logistic Regression was implemented as a baseline model to establish a reference for classification performance. As a linear classifier, it models the probability of a pixel belonging to the burned class using a logistic function applied to a weighted combination of input features. Despite its simplicity, this model provides interpretability by allowing examination of feature coefficients and their relative importance. However, its ability to capture complex nonlinear relationships in spectral data is limited.

Random Forest

The Random Forest model served as the primary traditional machine learning approach. It is an ensemble method that constructs multiple decision trees using bootstrapped subsets of the data and aggregates their predictions through majority voting. Random Forest is well-suited for this task due to its robustness to noise, resistance to overfitting, and ability to model nonlinear interactions between spectral features. It also provides implicit feature importance measures, offering insight into which spectral bands and indices contribute most to classification.

Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron is a feedforward artificial neural network capable of learning complex nonlinear mappings between inputs and outputs. The MLP used in this study consists of multiple fully connected layers with nonlinear activation functions. Compared to Logistic Regression, the MLP can better capture interactions between spectral features; however, it operates on individual pixels without explicitly incorporating spatial context. Model performance depends on architectural choices such as the number of hidden layers, neurons, and regularization techniques.

Convolutional Neural Network (CNN)

The Convolutional Neural Network extends the analysis by incorporating spatial information through localized receptive fields. Instead of classifying individual pixels independently, the CNN processes small image patches, allowing it to learn spatial patterns such as burn scars, edges, and texture differences. The architecture includes convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. This spatial awareness provides a key advantage over pixel-based models, particularly in heterogeneous landscapes.

Evaluation Metrics

Model performance was evaluated using three metrics:

Accuracy: Measures the overall proportion of correctly classified pixels but is misleading with class imbalance.

F1 Score: Balanced measure of classification performance, particularly for the burned class.

Cohen’s Kappa: Useful for imbalanced datasets, offering a more robust assessment of model reliability.

The performance of all four models across these metrics is summarized in Table 1 below. Collectively, these evaluation measures provide a comprehensive comparison of model effectiveness in detecting burned areas under varying data distributions.

Table 1: comparison on four models

	Metric		
	Accuracy	F1 Score	Cohen’s Kappa
Logistic Regression	0.978	0.883	0.871
Random Forest	0.978	0.884	0.872
MLP	0.997	0.985	0.984
CNN	0.993	0.949	0.945

The two best models, based on F1 score and Cohen’s Kappa, were the MLP and CNN. The accuracy of both exceeded 99%, and the F1 score and Cohen’s Kappa exceeded 90%.

The main drivers for each of the models is shown below:

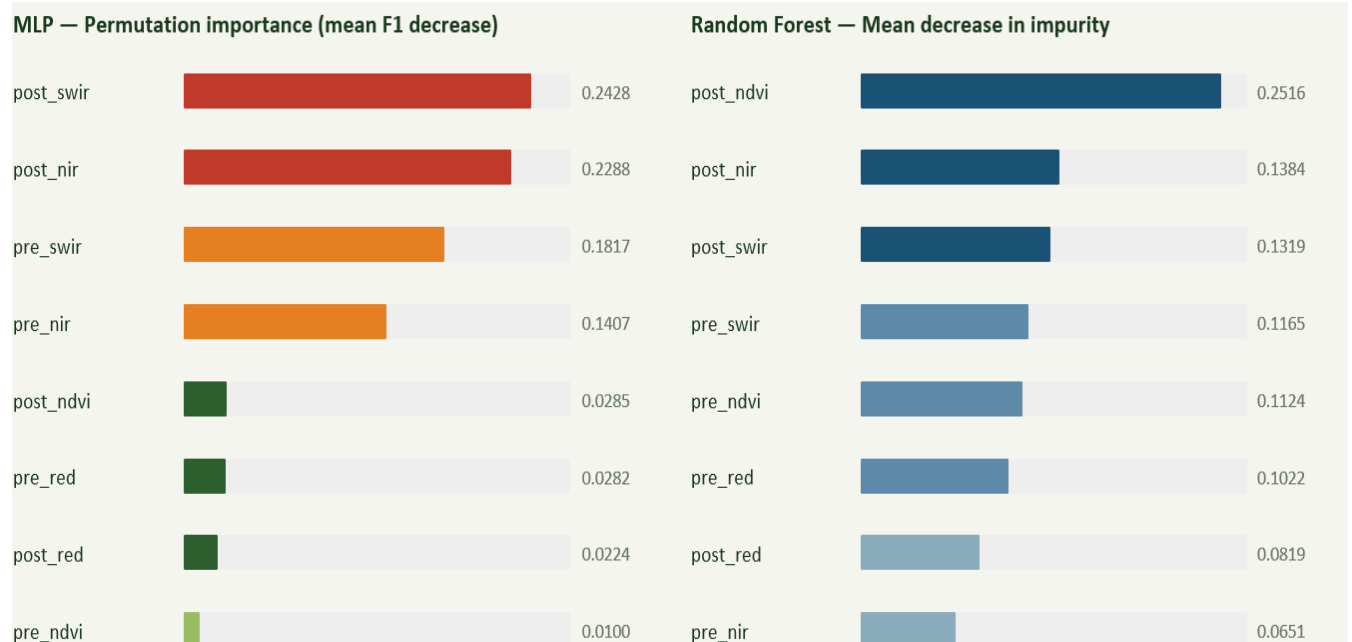


Figure 2: Main drivers of top models

Results

The performance of the developed models was evaluated based on their ability to accurately classify burned and unburned areas and to derive meaningful burn severity patterns across the study area. In addition to quantitative evaluation metrics, spatial outputs were analyzed to assess the quality and interpretability of the classifications.

Burn Severity Mapping

The dNBR was used to generate a burn severity map. Figure 3 shows both the continuous dNBR values and the corresponding classified burn severity categories. The classification was performed using standard dNBR thresholds, segmenting the landscape into unburned, low, moderate, and high severity classes. Figure 4 shows the distribution of burn severity.

The dNBR image clearly highlights areas affected by the fire, with higher values corresponding to greater vegetation loss and surface alteration. Regions classified as high severity exhibit strong spectral changes between pre-fire and post-fire conditions, particularly in the NIR and SWIR bands. Moderate severity areas form transitional zones surrounding the most intensely burned regions, while low severity areas indicate partial vegetation damage or surface disturbance.

2024 Jasper Wildfire — Burn Severity Assessment

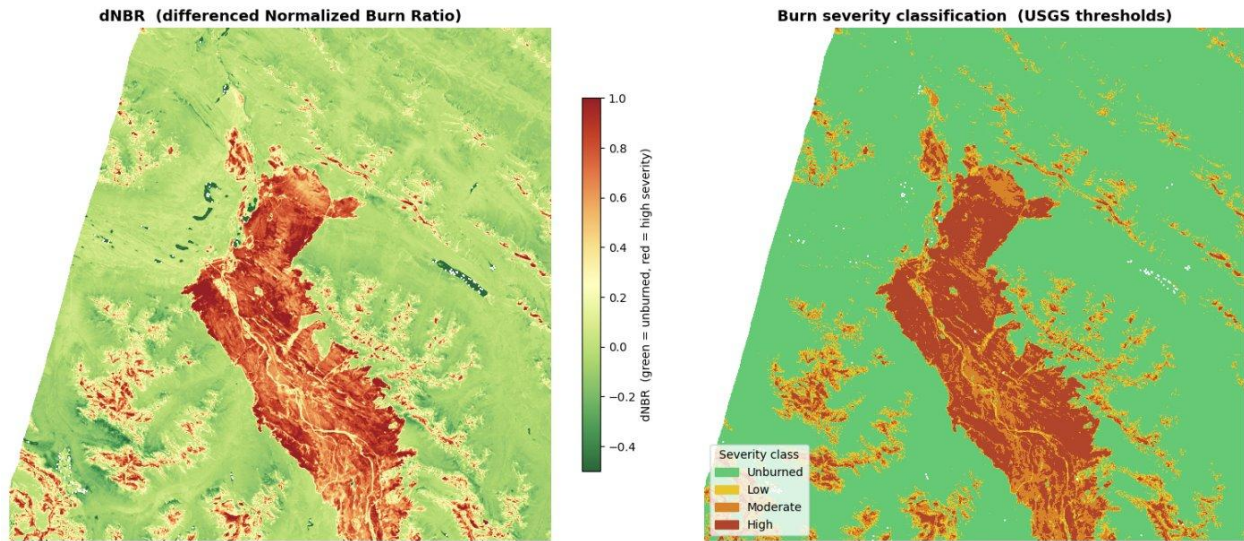


Figure 3: dNBR and burn severity classification

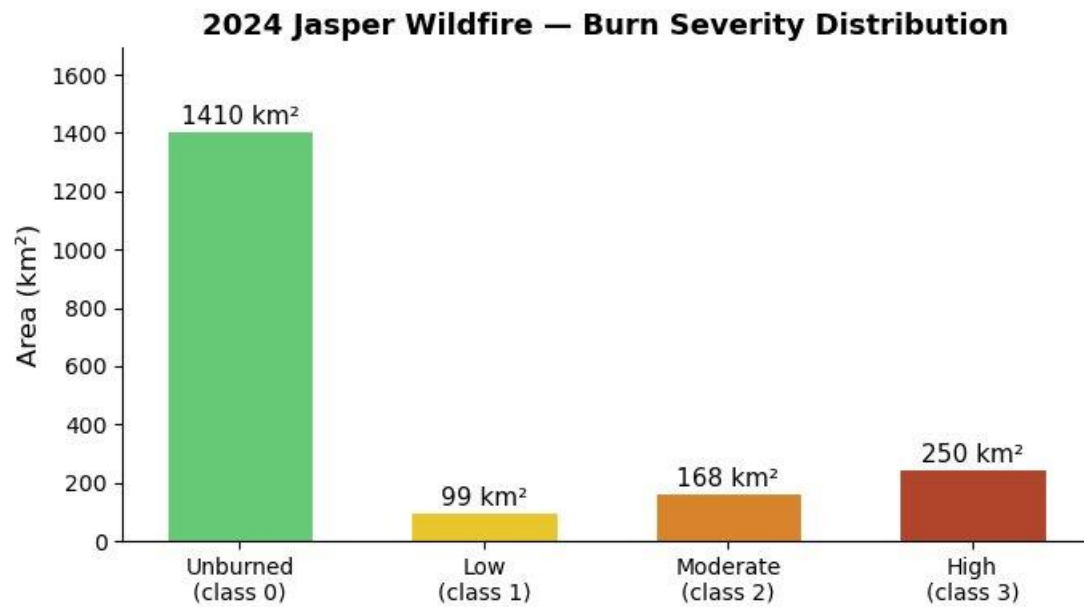


Figure 4: Burn Severity Distribution

Model Performance Overview

Across all models, differences in classification performance were observed, particularly in handling the imbalanced test dataset. Simpler models such as Logistic Regression showed limitations in capturing nonlinear spectral relationships, resulting in lower F1 scores and Cohen's Kappa values. The Random Forest model demonstrated strong performance, effectively balancing precision and recall while maintaining robustness to noise.

The Multi-Layer Perceptron (MLP) improved upon the baseline by capturing nonlinear feature interactions, though its lack of spatial awareness limited its ability to fully distinguish complex burn patterns. The Convolutional Neural Network (CNN) achieved the best overall performance, benefiting from its ability to incorporate spatial context and learn localized patterns associated with burn scars.

Overall, the results demonstrate that incorporating both spectral and spatial information leads to more accurate and reliable burned area classification, with the CNN providing the most consistent performance across evaluation metrics and spatial outputs.

Conclusion

This study demonstrates the effectiveness of combining satellite imagery with machine learning and deep learning techniques to map burned areas and assess fire severity. Using multispectral Sentinel-2 data, spectral indices such as NDVI, NBR, and dNBR were computed to capture vegetation changes and fire-induced landscape alterations. Logistic Regression, Random Forest, Multi-Layer Perceptron, and Convolutional Neural Network were evaluated, with the CNN achieving the highest accuracy and best handling of spatial patterns in burn scars.

The spatial outputs, including burn severity maps and severity distributions, revealed a heterogeneous fire pattern in the 2024 Jasper Wildfire Complex, with clusters of high, moderate, and low severity reflecting variations in fuel, topography, and fire behavior. The total burned area and severity proportions provide valuable information for post-fire ecological assessment and recovery planning.

This work highlights the potential of integrating remote sensing and advanced computational models for rapid, large-scale wildfire assessment. The methodology can be applied to other regions and fire events, supporting informed decision-making for disaster management, ecological monitoring, and resource allocation in wildfire-prone landscapes.

References

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